

LESSON GUIDE

Technology Plus

Unit 3: Hardware

2.1.1 – Different Computing Devices

Lesson Overview:

In this lesson, students will explore the purpose, features, and uses of different computing devices.

Students will:

- Explain the purpose of different computing devices.
- Identify the key features and common uses of different computing devices.
- Evaluate a scenario and determine which computing device would meet a user's needs.

Guiding Question: What are the common types of computing devices and what are their purposes?

Suggested Grade Levels: 8-12

Technology Needed: No

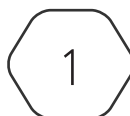
Approximate Time Required: 55-80 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 2.1 – Explain common computing devices and their purposes.
 - o Smartphones
 - o Tablets
 - o E-readers
 - o Laptops
 - o Workstations
 - o Servers
 - o Gaming consoles
 - o Virtual reality systems
 - o Augmented reality systems

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Lesson Background

Background Information:

Students should already be familiar with the basics of computing from lesson 1.1.1 and internal computing components from lesson 2.2.1. The introduction to this lesson prompts students to remember what makes a device a computer. This lesson covers different computing devices in sections by mobile devices, computers, and gaming consoles. Students are likely already familiar with these devices but not their purpose and how they compare to each other. Students will identify the purpose, key features, and common uses of each computing device - allowing them to recognize which device would be valuable in distinct situations.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 2.1.1 – Different Computing Devices • Activity Slides 2.1.1 – Computing Devices Bingo • Activity 2.1.1 – Bingo Strips (cut out)
Materials per Student	• Student Notes 2.1.1 – Different Computing Devices • Student Handout 2.1.1 – Different Computing Devices • Activity 2.1.1 – Bingo Cards (one each) • Bingo Chips or something similar • Assessment 2.1.1 – Different Computing Devices (optional)

2.1.1 – Different Computing Devices

Guiding Question: What are the common types of computing devices and what are their purposes?

Lesson Launch (5 minutes)

Is It a Computer?

- Display the images on slide three.
- **Ask:** Which of these devices are computers? Allow think time.
- **Optional Follow-up Questions:** What makes something a computer? Is it the shape or size? The screen? Or what it can do?

All of the images are computers; a computer is an electronic device that can receive input, process data, store information, and output results. All computing devices have a motherboard, a processor, and storage.

- We often think of computers as laptops or desktops, but computing devices go far beyond that. A computer is an electronic device that can receive input, process data, store information, and output results. All computing devices have a motherboard, a processor, and storage.
- We use computers for varying tasks, so it is essential to have various types of computers. Choosing a computer based on our needs or wants improves our experience with technology and creates convenience in our daily lives.
- Tell students that today they will explore common types of computing devices and their purposes.

From Pocket to Powerhouse (30-40 minutes)

- Distribute Student Handout 2.1.1 – Different Computing Devices and encourage students to fill in the answers during the lesson.

Mobile Devices

- Introduce mobile devices, also called hand-held devices, and the common types of mobile devices that will be discussed in the lesson.
 - Mention mobile devices have specific characteristics to set them apart from computers, such as touchscreen interface, built-in GPS, vibration features, and cellular connectivity.
- Discuss the key features of mobile devices:
 - Touchscreen interface allows easy navigation and interaction with apps, media, and functions.
 - The OS of a mobile device manages the hardware and determines the range of apps and features available on the specific device. A device's OS cannot be changed.
 - Apps and software available depend on the OS and can be downloaded from the device's app store. E-readers come with an integrated store or library that allows readers to purchase books.

- Most mobile devices can connect to the internet via Wi-Fi; some smartphones and tablets can connect to cellular networks.
- Mobile devices have a thin, lightweight design for portability.
- Mobile devices have rechargeable batteries that are often long-lasting but can vary in longevity based on the model and usage.
- Discuss the common uses of mobile devices:
 - Smartphones and tablets provide easy access to websites, social media, and other online services.
 - Smartphones and tablets can be used to watch videos, listen to music, read articles, or books.
 - Smartphones or tablets can be used to manage schedules and email, take notes, or use office collaboration apps.
 - Tablets can be used for interactive learning, e-books, and educational apps. E-readers can be used for reading textbooks, research papers, and other academic materials.
- Students should note that all key features listed under mobile devices apply to smartphones, tablets, and e-readers.

Smartphones

- Introduce smartphones as a type of mobile device, their key features and common uses.
 - Mention that smartphones have become essential to daily life and offer convenience and connectivity in a compact, portable form.
- Discuss the key features of smartphones:
 - Smartphones have high-resolution cameras with advanced features for taking photos and videos in any environment.
 - Smartphones have GPS functionality and location-based services like navigation, location tracking, and location-based recommendations.
 - Smartphones have a range of sensors to enhance the functionality of apps and provide accessibility features. Some examples include fingerprint unlock, face recognition unlock, temperature check, ambient light sensor (for auto-brightness), and an accelerometer (for auto-rotate or step counting).
- Discuss the common uses of smartphones:
 - Making calls, sending texts, and video-calling.
 - Using maps or GPS services for directions or location tracking.
 - Capturing high-quality photos and videos that can be easily shared.

Tablets

- Introduce tablets as a type of mobile device, their key features and common uses.
 - Mention tablets are suitable for personal and professional use due to the flexible and portable computing experience they offer.

- Discuss the key features of tablets:
 - Tablets have high-resolution screens, larger than smartphones but smaller than laptops - in most cases. The largest tablets are around the size of the smallest laptops - 13 inches. Tablets are still more portable due to only being a screen.
 - Tablets can be paired with accessories like keyboards, styluses, and docking stations.
- Discuss the common uses of tablets:
 - Many people use tablets for drawing, graphic design, and other creative tasks especially those with stylus support.

E-readers

- Introduce e-readers as a type of mobile device, their key features, and common uses.
 - Also known as electronic readers.
 - Mention that e-readers are not optimal for playing games or watching videos.
- Discuss the key features of e-readers:
 - E-readers use e-ink technology to mimic ink on paper and reduce eye strain making reading easier in bright sunlight.
 - Users can easily customize text size, font style, and line spacing for readability and comfort.
 - E-readers have large storage to hold thousands of e-books at once and some models support SD cards for expandable memory.
 - Users can highlight text, make notes, and bookmark pages like a physical book.
- Discuss the common uses of e-readers:
 - Reading digital books provides a way to carry a large collection of books in one device.
 - Traveling and wanting to avoid the physical bulk of carrying multiple books.
 - Accessibility features like text-to-speech and adjustable text size for every book.

Computers

- Introduce computers and the common types that will be discussed in the lesson.
 - **Every device in the lesson is a computer. Here “computer” is set apart as different than mobile devices based on their connection options.**
 - Mention that computers typically include ports that allow users to connect peripherals such as a mouse, keyboard, speakers, USB hard drives, and printers to the motherboard.
- Discuss the key features of computers:
 - A laptop’s OS choice impacts software compatibility and user experience. Workstations typically need a more professional OS for advanced features and better support for business applications. Servers need an OS that provides robust network and security features.
 - All computers are built with processors, memory, and storage capabilities that provide varying levels of performance. Higher-end model laptops can handle gaming, editing, and programming

tasks. Workstations are built to handle complex computations and demanding applications. Servers are built to handle multiple simultaneous requests and provide reliable, uninterrupted service.

- Workstations and servers are built for reliability and performance, they often include features like error-correcting code memory, redundant power options and advanced cooling systems to ensure they can provide uninterrupted service.
- Workstations offer options like high-capacity hard drives and solid-state drives; they also support redundant array of independent disks (RAID) configurations. Servers also use RAID configurations and support network-attached storage, or storage area networks.
- Workstations have enhanced networking capabilities when paired with other professional equipment and networks. Servers have advanced networking features to handle large amounts of data traffic.

Laptops

- Introduce laptops as a type of computer with multiple connection options, their key features, and common uses.
 - Laptops are sometimes referred to as notebook computers or hybrids. A hybrid laptop has a touchscreen interface with all other components of a traditional laptop.
 - Mention that laptops can be harder to fix than a bigger computer due to their small, compact components.
- Discuss the key features of laptops:
 - Laptops have built-in components like a screen, keyboard, touchpad or trackpad, and speakers making them convenient for portable use.
 - Laptops are larger and heavier than tablets, but they are still portable and can be carried in bags or briefcases.
 - Laptops offer multiple connection options like Bluetooth, Wi-Fi, USB ports, HDMI outputs, and sometimes Ethernet ports.
 - Laptops have rechargeable batteries that provide power away from a stationary outlet.
 - Screens vary in size, resolution, and quality.
 - Laptops support a variety of applications for their user's possible needs.
- Discuss the common uses of laptops:
 - Laptops are essential to students and professionals completing assignments and projects.
 - Users can watch movies, play games, listen to music, or browse the web. Laptops offer higher video and audio quality in comparison to mobile devices.
 - Laptops are commonly used by professionals in creative work such as graphic design, video editing, and music production.
 - Laptops offer flexibility to work or learn from various locations.

Workstations

- Introduce workstations as a type of computer with multiple connection options, their key features, and common uses.
 - Workstations are a type of desktop. They are optimized for tasks that need a generous amount of computing power and precision.
 - Workstations connect their devices like screens, keyboards, mice, and speakers through ports.
 - Although workstations are high-performance, they can still be utilized as everyday computers for office or personal tasks.
 - Some workstations connect directly to a local area network (LAN) and communicate with servers to use a network's resources.
- Discuss the key features of workstations:
 - Workstations use professional grade graphics cards designed for tasks like 3D rendering, computer-aided design (CAD), and video editing.
 - Most workstations allow users to upgrade and customize their system as needed.
 - Users can pair special input devices to enhance productivity and accuracy in tasks like graphic tablets and high-precision mice.
- Discuss the common uses of workstations:
 - Workstations are used for CAD applications, simulation, and 3D modeling in fields like architecture, engineering, and industrial design.
 - Creative professionals use workstations to produce high-definition videos, audio mixing, and complex visual effects.
 - Researchers use workstations for data analysis, simulations, and complex computations.
 - IT professionals use workstations for coding, testing, and running complex application and development environments.

Servers

- Introduce servers as a type of computer with multiple connection options, their key features, and common uses.
 - Mention there are two distinctions when categorizing servers: dedicated and nondedicated.
 - Dedicated servers are assigned to provide specific applications or services for a network and require fewer resources for efficiency.
 - Nondedicated servers are expected to have more flexibility for day-to-day use because they are assigned to handle one or more network services and local access for a user.
- Discuss the key features of servers:
 - Servers can be scaled up by adding more resources to a server or scaled out by adding more servers to a network.

- Servers usually have redundant components such as multiple power supplies, cooling fans, and network connections to prevent downtime.
- Servers come with tools that allow administrators to oversee performance, manage configurations, and perform maintenance tasks remotely.
- Servers have enhanced security features such as firewall capabilities, encryption, and access controls.
- Discuss the common uses of servers:
 - Servers host websites and manage pages and traffic for users accessing sites over the internet.
 - Servers manage email communications for users within an organization or across the internet.
 - Servers provide centralized storage for files and data allowing users to access and share information across a network.
 - Servers host databases, handle queries and transactions from applications and users.
 - Servers run applications and provide services such as enterprise resource planning systems, customer relationship management systems, and other business-critical applications.
 - Servers host virtual machines by using virtualization technologies, allowing multiple virtual servers to run on a single physical server.
- Students should recognize that workstations are clients on the network and servers manage the network. Servers and workstations are differentiated by what OS it has installed and the role it plays on the network.

Gaming Consoles

- Introduce gaming consoles, their key features, and common uses.
 - Mention that gaming consoles also use virtual and augmented reality to mix real-world views with computer graphics and provide enhanced realistic experiences.
- Discuss the key features of gaming consoles:
 - Gaming consoles are built with specialized hardware for gaming including powerful processors and graphics cards.
 - Gaming consoles use controllers or gamepads for input.
 - Most consoles feature exclusive games that are developed for a specific platform.
 - Some consoles have integrated online services and stores where users can purchase, download, play games, and interact with other gamers.
 - Some consoles allow users to play games from previous generations of the same console family.
 - Some consoles support virtual reality or augmented reality.
- Discuss the common uses of gaming consoles:

- Playing video games.
- Accessing streaming services, digital media playback, and integration with home entertainment systems.
- Local or online multiplayer use for communication and competitive play.

Virtual Reality (VR) Systems

- Introduce VR systems, their key features, and common uses.
- Discuss the key features of VR systems:
 - The head-mounted display (HMD) is the device worn on the user's head that displays the VR environment.
 - Most VR systems come with motion controllers that track the user's movements and allow interaction with the virtual environment.
 - Tracking sensors or cameras are used to accurately track a user's head and body movement, allowing the virtual environment to respond to the user's actions.
 - VR systems require powerful computing hardware to provide high-quality graphics and maintain a high frame rate.
 - Many HMDs include built-in headphones to enhance virtual realism.
- Discuss the common uses of VR systems:
 - VR systems are widely used in gaming for immersive experiences.
 - Some fields like aviation, medicine, and the military use VR for training and simulations.
 - Some VR applications provide educational experiences by simulating historical events, scientific phenomena, and complex systems.
 - Some VR platforms offer social spaces for user interaction.

Augmented Reality (AR) Systems

- Introduce AR systems, their key features, and common uses.
- Discuss the key features of AR systems:
 - AR glasses or headsets can be worn on the head and display digital information over the real-world.
 - Some computing devices have AR software that allows the device to capture the real-world while the software overlays digital information.
 - AR systems use cameras and sensors to capture a user's environment and accurately place digital objects for interaction.
 - AR systems need processing power in real-time to handle image and sensor data and integrate digital content with the physical world.
 - AR systems rely on internet connectivity to access cloud-based data.
- Discuss the common uses of AR systems:

- AR is used in gaming and entertainment to blend digital elements with the real-world view.
- AR can enhance navigation by overlaying directions and information on the real-world view.
- AR applications can assist shoppers by allowing virtual try-ons.
- AR can provide interactive learning experiences.
- AR can assist maintenance technicians by overlaying instructions or diagrams on equipment when performing repairs.
- AR can be used for medical training, surgeries, and patient care.
- AR filters and effects are commonly used on social media platforms to add virtual elements to photos and videos.

Activity: Computing Device Bingo (15-25 minutes)

- Each student should have one bingo card and bingo chips or something similar to mark their spaces during the game.
- Each bingo card is a 3X3 grid, with 8 answer choices and one FREE space.
- The bingo strips should be cut out and folded up or placed into a bag where the scenario on them is not seen. There are 11 scenarios total.
- The amount of bingo chips per student depends on the bingo game you choose to play.
- The caller will pull a scenario, read it aloud, and wait for students to mark it on their bingo card if needed.
- If a user calls bingo, read the called scenario's again and have them call back what they marked off to check their answers.
- Students must call bingo and have correct answers to win the game.
- You can extend the game to multiple rounds by changing the winning rules and having students swap bingo cards.

Winning Rules (Choose one per round):

1. Straight Line: Student marks three devices in a row (horizontal, vertical, or diagonal).
2. Blackout: Student marks all nine spaces.
3. Corners: Student marks all four corners.
4. X Pattern: Student marks both diagonals and forms an X on their card.

There are "Winning Rule" slides, feel free to display a slide based on which rules you pick, or stay on the "Directions" slides.

Note: The answer to each scenario is below. Do not read from this during the game, please pull bingo strips at random.

- A user wants a type of computing device that is portable, allows cellular connectivity, and a

rechargeable battery. **Mobile devices**

- A user wants a type of computing device that uses ports as connection options to enhance their functionality. **Computers**
- A user wants a device for gaming that will allow them to interact with their friends online. **Gaming console**
- A user wants a device for communication, GPS functionality, and a high-quality camera. **Smartphones**
- A user wants a device that is portable, has a touchscreen, and makes creative tasks easy. **Tablets**
- A user wants a device that will allow them to read books digitally without needing to worry about lighting. **E-readers**
- A user wants a device that has integrated components for portability, has the power of a desktop, and is great for remote learning. **Laptops**
- A user wants a device that has high performance features for their work in video production while maintaining expandability for their personal needs. **Workstations**
- A user wants a device that will allow them to manage an organization's email communications and storage access to files and data. **Servers**
- A user wants a device that will allow them to take a virtual tour of the Grand Canyon. **VR systems**
- A user wants a device with software that will allow them to try on clothes and shoes at home. **AR systems**

Putting It All Together (5-10 minutes)

- Review what students learned today by discussing the questions below as a class.
- What are some key features of mobile devices?

Key features of mobile devices include touchscreen interface, operating systems, apps and software, cellular connectivity, portability, and long-lasting battery.

- What are some key features of computers?

Key features of computers include operating systems, performance, reliability, storage, networking, and connectivity.

- What are some key features of gaming consoles?

Key features of gaming consoles are dedicated hardware, integrated controllers, exclusive games, online services and stores, backward compatibility, and VR and AR support.

- What are some benefits of using a tablet over a smartphone?

Tablets have a bigger display and can sync with accessories like keyboards and styluses to make the tablet more "laptop like" than a smartphone.

- What are some benefits of using a workstation over a laptop?

A workstation is designed for higher performance and may have professional graphics cards to support users that need a generous amount of computing power for their tasks. Workstations are

also great for users who don't need portability.

- What is the difference between AR and VR systems?

AR systems put digital content over the real-world and VR systems bring a user into the digital world.